CANCER CARE · CLINICIAN FOCUSED

RoundsAI

Oncology Ward Round Copilot

Turn a 20-minute oncology rounding encounter into a structured, evidence-linked clinical note in under 90 seconds — with 100% on-device voice privacy and ₹0 cloud API cost.

TEAM NAME
Binary Minds (Code: 1RKUPH)

DOCTOR PARTNER & CLINICAL LEAD
Dr. Amrut S Kadam · Victoria Hospital, BMCRI

TECHNICAL LEADS
Surya Thejas & Sai Santhosh

CHALLENGE TRACK
Doctor Productivity & Knowledge Assistant

THE PROBLEM

The Oncology Ward Rounding Crisis

High cognitive load, fragmented diagnostic data, and severe physician documentation burnout.

Extreme Cognitive Load

Oncologists round on 15–20 complex cancer patients daily, balancing verbal symptoms, printed CBC lab sheets, and multi-agent chemotherapy cycles under strict time constraints.

15–20 PATIENTS EVALUATED PER MORNING

The Documentation Tax

Clinicians spend 2 to 3 hours every evening manually typing progress notes, calculating lab deltas, and cross-referencing guideline dosages into clunky hospital EHRs.

2–3 Hrs LOST TO NIGHTLY DATA ENTRY

Generic AI Scribes Fail

Commercial voice tools constantly butcher specialized cancer regimens (hearing "fall fox" instead of mFOLFOX6) while transmitting patient audio to cloud servers, risking HIPAA breaches.

Zero ONCOLOGY DOMAIN COMPREHENSION

THE SOLUTION

RoundsAI: Built for the Hospital Ward

A connected clinical copilot harmonizing bedside mobile capture with desktop assessment.

Bedside Mobile Capture

Doctors don't carry laptops between beds. Our responsive mobile companion allows natural voice dictation and camera photo scanning of paper lab reports right at the bedside.

- ✓ Local offline Whisper AI voice engine
- ✓ 1-tap camera lab sheet photograph
- ✓ Instant bedside patient context selector

Live Bedside-to-Desktop Sync

One tap transmits the bedside round. The nursing station workstation detects incoming data in real-time, loading findings with zero manual re-typing.

- ✓ Real-time background polling sentinel
- ✓ 1-click injection into Assessment Station
- ✓ Zero copy-pasting or file transferring

100% On-Device Privacy

Runs on local CPU with faster-whisper. Sensitive patient voice recordings never leave the hospital network, ensuring zero data leakage and ₹0 recurring API fees.

- ✓ 120+ chemotherapy phonetic normalizer
- ✓ Standardizes units (ANC 900 /µL, WBC 2,100)
- ✓ Zero cloud audio transmission

SYSTEM ARCHITECTURE

4-Tier Edge-Assisted Architecture

Designed for low latency, zero external audio leakage, and hospital resilience.

1

Client Tier

Next.js 15 (React 19)

- ✓ Bedside Mobile Companion (`/mobile`)
- ✓ Desktop Station (`/`)
- ✓ Longitudinal Analytics (`/analytics`)
- ✓ Hospital vector SVG UI (0 emojis)

2

Local Audio AI

faster-whisper (int8)

- ✓ 100% On-device CPU inference
- ✓ Oncology lexicon prompt priming
- ✓ 120+ phonetic regex normalizer ("fall fox" → mFOLFOX6)
- ✓ Sub-second local transcription

3

Clinical Brain

FastAPI Application

- ✓ Structured lab entity extraction
- ✓ CTCAE v5.0 baseline comparisons
- ✓ Automated SOAP note assembler
- ✓ Curated ASCO / NCCN guideline RAG

4

Persistence

Storage & Audit

- ✓ Dynamic round memory cache
- ✓ SQLite & Supabase PostgreSQL
- ✓ Cycle-over-cycle trajectory store
- ✓ Immutable clinical audit trail

PRODUCT CAPABILITIES

End-to-End Clinical Experience

From bedside dictation to verified documentation and cycle trajectory tracking.

A

Bedside Dictation & Scan

Oncologist dictates naturally: *"Patient 104 developed fever 38.4°C, WBC 2100, ANC 900."* The system standardizes terms and snaps paper lab sheets with 1-tap confirmation.

30 Sec BEDSIDE ENCOUNTER CAPTURE

B

Live Sync & Note Drafting

Workstation detects transmitted round. 1-click loading auto-fills findings, compares labs against baseline, and generates a standardized oncology SOAP note for physician sign-off.

1-Click ZERO MANUAL TYPING REQUIRED

C

Longitudinal Analytics

Plots ANC and platelet nadir curves across chemotherapy cycles (Cycle 1, 2, 3), tracking immune recovery and displaying past round audit histories.

Full History CYCLE-BY-CYCLE TRAJECTORY

NOVELTY & GUARDRAILS

Why RoundsAI Wins & Stays Compliant

Purpose-built domain innovation designed strictly within ethical healthcare scope boundaries.

🏆 Core Technological Novelty

- ✓ **Hardware Harmony:** Phone acts as the portable bedside sensor; computer serves as the high-density review station.
- ✓ **Zero-Cloud Audio Privacy & ₹0 Cost:** On-device Whisper AI ensures sensitive patient voice never leaves the hospital firewall.
- ✓ **Oncology-Specific Normalizer:** Tailored regex phonetics for 120+ chemotherapy regimens and lab counts.
- ✓ **Instant Bedside-to-Desktop Sync:** Seamless background transmission without manual file sharing or email.

🛡️ Strict Scope Guardrail Compliance

- ✓ **Assistive Documentation Only:** Does not provide autonomous diagnosis, clinical risk scoring, or unverified treatment orders.
- ✓ **Objective Data Presentation:** Visualizes historical lab trends and indexes published guidelines for clinician reference.
- ✓ **Mandatory Clinician-in-the-Loop:** The attending physician retains 100% clinical authority to review, edit, and sign every note.
- ✓ **Audit Provenance:** Every extracted metric links back to its documented transcript or paper lab source.

TEAM & IMPACT

Team Binary Minds & Measurable Value

Clinical grounding, technical depth, and proven productivity savings.

👥 Interdisciplinary Team

Dr. Amrut S Kadam · Clinical Lead
Victoria Hospital, Bangalore Medical College & Research Institute

Chalicheemala Surya Thejas · Technical & Team Lead
Full-Stack & Systems Architecture (suryathejas970@gmail.com)

Gaddam Sai Santhosh · Core Developer
AI & Frontend Engineering (saisanthosh05012006@gmail.com)

📈 Measurable Hospital Impact

70%+ DOCUMENTATION TIME REDUCED	**2–3 Hrs** SAVED PER PHYSICIAN DAILY
₹0 CLOUD VOICE TRANSCRIPTION FEES	**< 90 Sec** BEDSIDE TO APPROVED NOTE